

Multiple Choice

1. **Answer:** A

Options: A) Temporary inconsistencies among views of a file by different machines can result; B) The file system is likely to be corrupted when a computer crashes; C) A much higher amount of network traffic results; D) Caching makes file migration impossible

Note: Local caching in distributed file systems can lead to temporary inconsistencies because different machines may access and modify their cached versions of a file independently, resulting in divergent views of the same data until synchronization occurs.

2. **Answer:** D

Options: A) $O(\log N)$; B) $O(N \log N)$; C) $O(\log N) + O(1)$; D) $O((\log N)^2)$

Note: The correct answer is $O((\log N)^2)$ because the recurrence relation involves a logarithmic term that accumulates at each recursive step, leading to a quadratic growth rate in the logarithmic scale as N increases.

3. **Answer:** C

Options: A) C, Ruby; B) Python, Ruby; C) C, C++; D) Tcl, C#

Note: C and C++ are known for providing low-level memory management capabilities that allow direct manipulation of memory, which can lead to buffer-overflow errors if bounds checking is not implemented.

4. **Answer:** A

Options: A) Local variables; B) Program code; C) Dynamically linked libraries; D) Global variables

Note: The stack is used for storing local variables because it operates on a last-in, first-out principle that efficiently manages memory for function calls and their associated local scopes.

5. **Answer:** B

Options: A) Finding a longest simple cycle in G ; B) Finding a shortest cycle in G ; C) Finding ALL spanning trees of G ; D) Finding a largest clique in G

Note: Finding a shortest cycle in an undirected graph can be solved in polynomial time using graph traversal algorithms like breadth-first search (BFS) to systematically explore all paths and determine the shortest cycle.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: A session symmetric key is specifically generated for a single communication

session to ensure security and confidentiality, preventing the risks associated with key reuse such as cryptographic attacks.

7. **Answer:** True

Options: A) True; B) False

Note: Message integrity involves mechanisms such as checksums and cryptographic hashes that verify that data sent over a network remains unchanged and intact upon arrival at its destination, thus ensuring that it has not been altered during transmission.

8. **Answer:** False

Options: A) True; B) False

Note: Windows 7 incorporates several security enhancements, such as Address Space Layout Randomization (ASLR) and Data Execution Prevention (DEP), which significantly reduce the risk of buffer-overflow vulnerabilities compared to traditional UNIX systems.

9. **Answer:** True

Options: A) True; B) False

Note: The AH Protocol uses hashing to provide source authentication and data integrity through message integrity checks, but it does not encrypt the data, thus failing to ensure privacy.

10. **Answer:** False

Options: A) True; B) False

Note: Church's thesis, also known as the Church-Turing thesis, was not proven by Alan Turing but rather is a hypothesis about the nature of computable functions that was independently proposed by Alonzo Church and Turing, with Turing's work on Turing machines providing evidence for the thesis rather than a definitive proof.

Long Form Response

11. **Answer:** `def remove_datatype(test_tuple, data_type): | return (res)`

Note: correct because it outlines the creation of a function that, when implemented, will effectively filter out elements of a specific data type from a tuple, demonstrating an understanding of function definition and tuple manipulation in Python.

12. **Answer:** `def num_position(text): | return m.start()`

Note: correct because it utilizes the 'start()' method from a match object to identify and return the starting position of a number found within the given string, which is essential for separating and printing both the numbers and their respective positions.

End of Answer Key